

The empirical recurrence for OEIS A251257: recovering a conjecture that is not written down

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Abstract

OEIS A251257 records “Empirical recurrence of order 49”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 256$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 49, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 49 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A251257 is “Number of $(n+1) \times (7+1)$ 0..1 arrays with every 2×2 subblock having a single 1 or two 1s on the same edge.”. It has offset 1 and begins

850, 10108, 69270, 836546, 6080510, 71459218, 539961360, 6151714502, ...

The entry states:

Empirical recurrence of order 49 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:13:16, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 256$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 512$ exact terms of the model returns order 49 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{49} c_i a(n-i),$$

c_1	1	c_{14}	434763711429	c_{27}	3481947375897501	c_{40}	−2906513796382309
c_2	253	c_{15}	87080982368	c_{28}	−19497341514265213	c_{41}	−1123851889658786
c_3	−167	c_{16}	−4517357900579	c_{29}	−7456767093083207	c_{42}	603949031691888
c_4	−26912	c_{17}	−1116697546271	c_{30}	30591317701689603	c_{43}	223176892626860
c_5	10806	c_{18}	35553910757022	c_{31}	12012033454815927	c_{44}	−76815822989838
c_6	1621200	c_{19}	10069384905142	c_{32}	−36078943001722744	c_{45}	−26515483716488
c_7	−332064	c_{20}	−213623344160226	c_{33}	−14423131529357239	c_{46}	5262776969440
c_8	−62578242	c_{21}	−66557981392524	c_{34}	31592674700836115	c_{47}	1647304326336
c_9	3525016	c_{22}	983991594550701	c_{35}	12742139499632686	c_{48}	−145699075200
c_{10}	1655727924	c_{23}	329165514424294	c_{36}	−20200254198748044	c_{49}	−39785679360
c_{11}	91048322	c_{24}	−3478804358977975	c_{37}	−8136256221715597		
c_{12}	−31306221085	c_{25}	−1229857273960789	c_{38}	9217812681576885		
c_{13}	−4324103835	c_{26}	9425999363946700	c_{39}	3663838865259868		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 49, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $49 + 4$ terms removed returns order 49 as well, so $\deg q_{\min} = 49 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $49 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 49 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A251257.
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